

INDEPENDENT ENGINEERING CASE STUDY

Formally verified RTL optimization

Executive summary - SHA-1 / VTR 45 nm pilot

**5.98% better composite PPA across 64 paired
implementation seeds**

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Evidence snapshot: 5 August 2026

Academic open-FPGA estimate. Not ASIC signoff, a commercial FPGA result, or manufactured silicon.

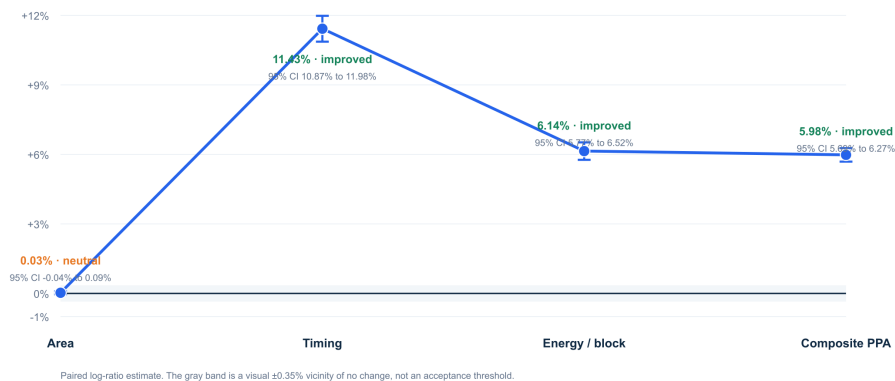
The decision

A four-line, cycle-equivalent RTL rewrite is accepted as the new champion: timing improves by 11.43%, workload energy by 6.14%, and total area remains statistically neutral.

Metric	Baseline median	Champion median	Paired result (95% CI)
Total area	16,614,693 MWTa	16,614,693 MWTa	0.03% better; -0.04% to +0.09% (neutral)
Critical path	15.0054 ns	13.28085 ns	11.43% better; 10.87% to 11.98%
Energy / block	12.3896 nJ	11.6399 nJ	6.14% better; 5.77% to 6.52%
Composite PPA	1.000000	0.940188	5.98% better; 5.69% to 6.27%

Certified PPA improvement vs. baseline

64 paired, fixed and search-disjoint VPR seeds - lower metric values are better



Paired log-ratio estimates over a fixed, search-disjoint 64-seed VPR pool. Positive values are improvements.

Why it is credible

Only sha.v was editable. NIST conformance and cycle tests passed before PPA; pinned EQY proved full sequential equivalence; and a 500-mutation qualification rejected all 482 functionally distinguishable cases. The transferable asset is the verification-first method: rank cheaply, then certify only formal-pass finalists on a fixed independent pool.

Claims boundary: this is an open academic FPGA estimate at VTR PTM45, not an ASIC or commercial-FPGA result. SHA-1 is used only as a legacy benchmark.